February 13, 2025

ASPEN AEROGELS

Q4 AND FULL YEAR 2024 FINANCIAL RESULTS CALL

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Q4 2024 Developments and Year-End Highlights

Record company revenue and continued gross margin expansion driven by focus on profitability and cash generation

Energy Industrial

- Delivered $53.1M in Q4 (~91% from EMF[1])
- Positioned for mid-teens CAGR through 2030

Thermal Barrier

- Delivered $70M in Q4, 32% YoY revenue growth
- 8th PyroThin® award and 3rd award in 2024

VOLVO

Operational Excellence

- 2024 gross margins of 40%
- Reduced OPEX to 2023 levels
- Diversified supply chain

Near-Term Focus Areas

- Maximizing capacity in Rhode Island facility
- Implementing capital-light and modular supply strategy

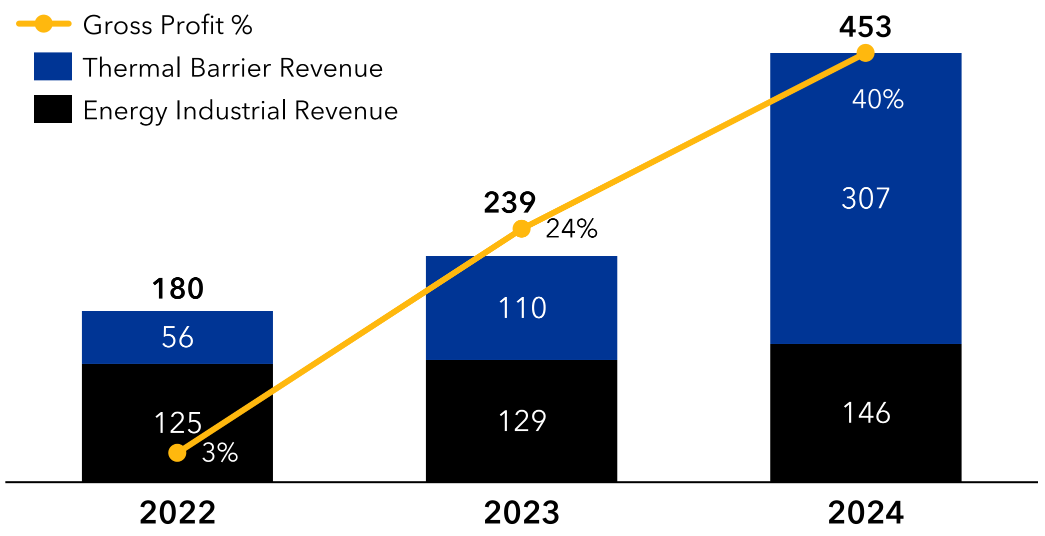
Balance Sheet and Cash Flow

- Ended FY24 with $220.9 million in cash
- Generated $20.9M of free cash flow in Q4
- Fully funded to execute current strategy and to explore opportunistic uses of capital

Financial Performance Progression

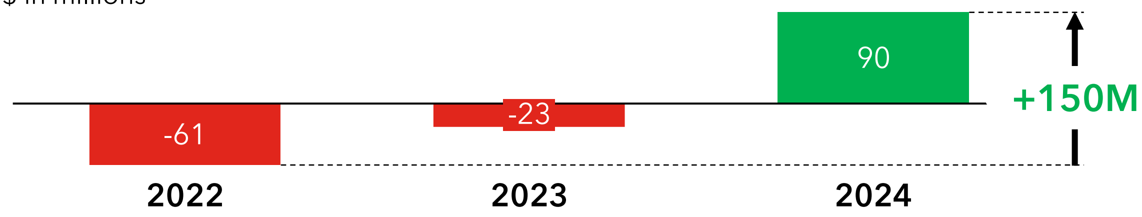
Revenue & Gross Margin %

$ in millions[2]



Adjusted EBITDA

$ in millions[2,3]



1-External manufacturing facility
2- Unaudited financials for 2024
3- See slide 9 herein for a reconciliation of net loss, the most directly comparable GAAP measure, to Adjusted EBITDA

Note: Figures may not total due to rounding

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Q4 and Full Year 2024 Financial Highlights

Demonstrating business model scalability while tightly managing fixed costs and reducing capital expenditures

$ in millions except per share figures	Q4 2023	Q4 2024[1]	2023	2024[1]	Commentary
Revenues *YoY Growth*	$84.3 *41%*	$123.1 *46%*	$238.8 *32%*	$452.7 *90%*	• $306.8M of Thermal Barrier revenues in 2024 led to 3x revenue growth YoY, headlined by GM's continued Ultium production ramp • Record Energy Industrial revenues of $53.1M in Q4 drove 13% revenue growth YoY to $145.9M in 2024
Net Inc. (Loss) *Margin*	$(0.5) *(1%)*	$11.4 *9%*	$(45.8) *(19%)*	$13.4 *3%*	• 5th Consecutive quarter with >35% gross margin, which led to a YoY improvement of 3x in Gross Profit for 2024 • 4% improvement in material costs YoY as a % of sales
Adj. EBITDA[2] *Margin*	$9.1 *11%*	$22.7 *18%*	$(22.9) *(10%)*	$89.9 *20%*	• Demonstrated profit potential through fixed cost absorption; total Cost of Goods Sold as a % of sales improved by 17% YoY • Gross profit improvement and careful OPEX management led to >50% of the $214M in YoY revenue growth falling through to EBITDA
EPS (Loss) Diluted	$(0.01)	$0.14	$(0.66)	$0.17	• Continued investments to deliver on increasing customer demand • Delivered positive EPS in Q2-24, Q4-24, and FY 2024
Cap Ex	$27.8	$14.8	$175.5	$86.3	• Invested $86.3M of CAPEX for 2024 in EV Thermal Barrier assembly equipment, the Statesboro Plant, and our existing East Providence Plant • Reduced CAPEX spend in 2024 by >50% YoY

1- Unaudited interim financials for 2024
2- See slide 9 herein for a reconciliation of net income (loss), the most directly comparable GAAP measure, to Adjusted EBITDA

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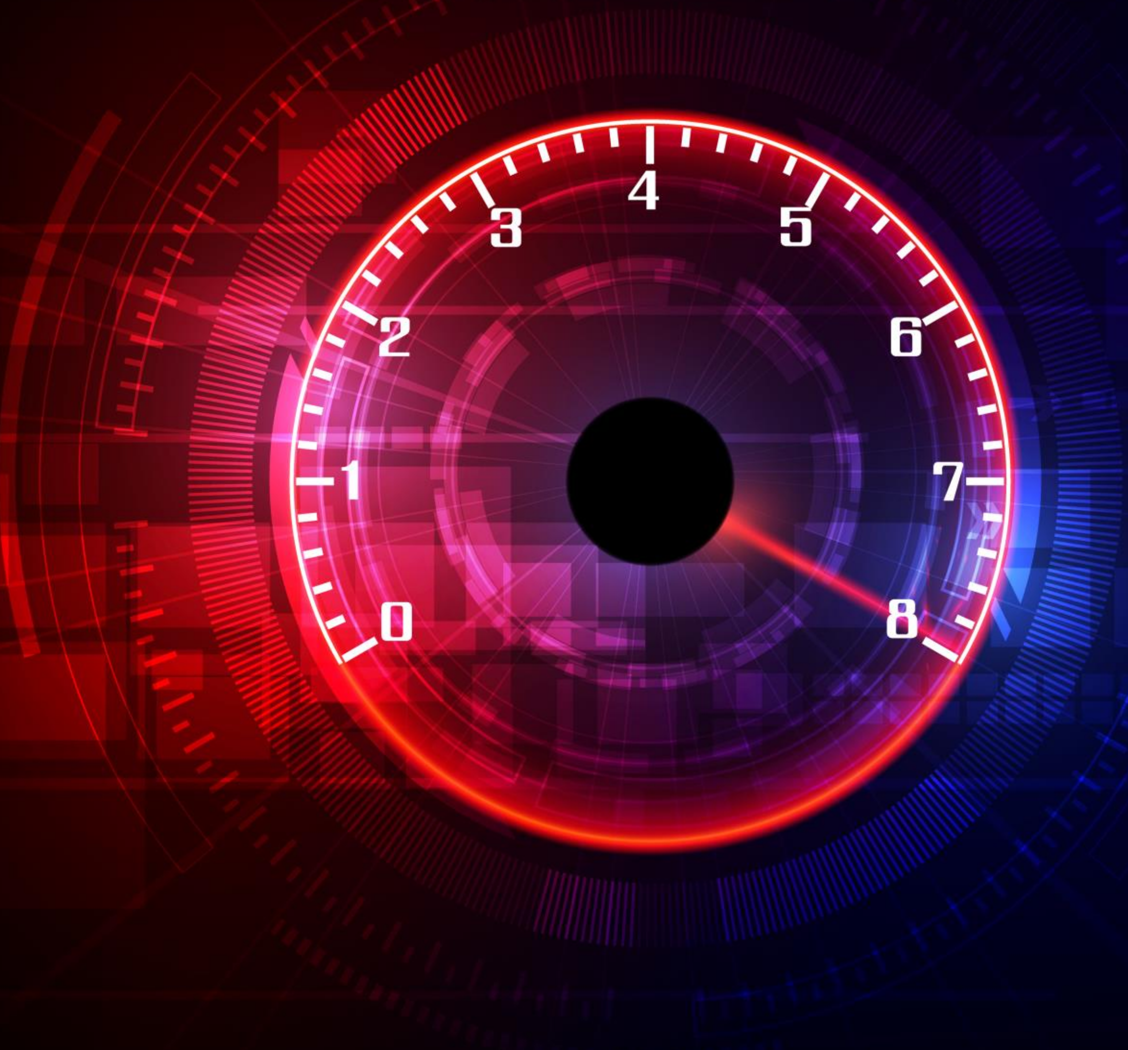
Business Segment Dynamics

Leveraging our Aerogel Technology Platform® into dynamic, complementary, and expanding end markets

	2024	2025
Energy Industrial	**Demand Drivers** • Strong demand due to geopolitics • LNG, power generation, and C02 recapture **Considerations** • Aspen supply constrained • 5-week turnaround at EMF in Q3 of 2024	**Demand Drivers** • Acceleration in new projects and permits • Middle East expansion **Considerations** • Active US import tariff management • Pricing and cost optimization
Thermal Barrier	**Demand Drivers** • Initial vehicle inventory buildup to ~70 days • New nameplate launches • Low initial OEM module / pack yields **Considerations** • High level of production incentives • Stringent near-term regulatory environment • High EV lease rates in the US for $7,500 credit	**Demand Drivers** • High vehicle inventory levels to start the year • Production likely directly linked to sales • European OEM cell supply volume **Considerations** • Potential regulatory environment relief • Effect of incentives and tax credits on sales • High interest rates and affordability questions

Driving Performance in 2025 and Beyond

We continue to regear the company to create value in a broad range of demand outcomes without additional capital



Minimized CAPEX Investments

- Leveraging modular external manufacturing supply model
- Maximizing Rhode Island capacity
- Avoiding $671M of indebtedness without Plant II

Optimized the Cost Structure

- Lowered annual fixed costs by over $35M
- Diversified raw material supply in advance of tariffs
- OPEX run rate brought back to 2023 levels

Strengthened the Balance Sheet

- Cash balance fully funds current strategy and plans
- Started paying down the term loan
- Potential for further capital structure optimization

2025 Q1 Outlook and Historical Trends

While a broad range of demand outcomes are possible, we remain focused on protecting profitability and capturing upside

Q1 2025 Outlook[2]

$ in millions, except per share figures

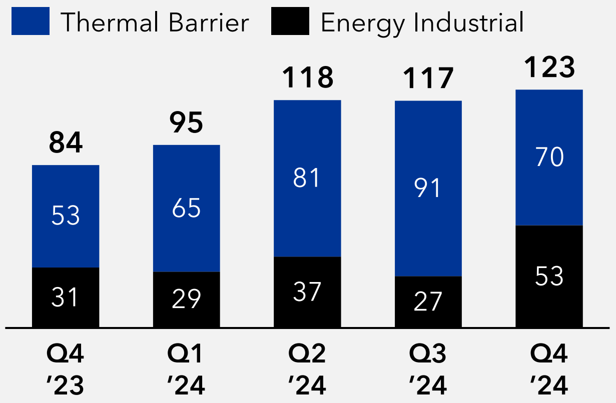
Revenues	$75 - $95
Net Income (Loss)	$(15) - $0
Adj. EBITDA[2]	$0 - $15
EPS (Loss)[3]	$(0.18) - $0.00
Cap Ex (excl. Plant II)	<$7

1- Unaudited interim financials for 2024
2- See slide 9 herein for a reconciliation of net income (loss), the most directly comparable GAAP measure, to Adjusted EBITDA for the presented period
3- Current outlook assumes diluted weighted average shares outstanding of 82.0 million for the quarter

Historical Financial Trajectory[1]

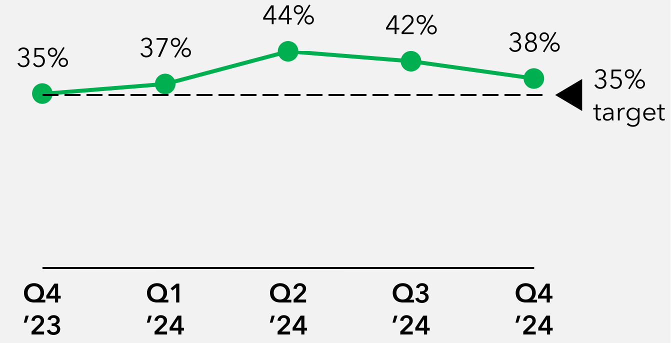
Revenues

$ in millions



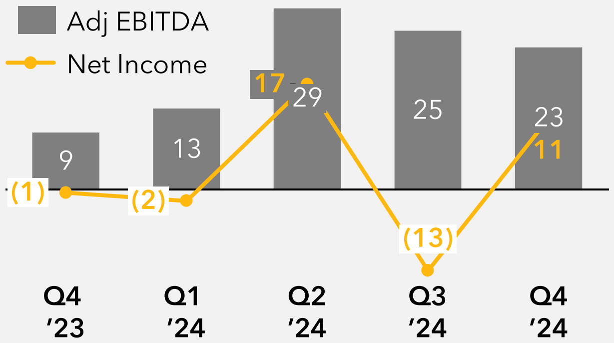
Gross Profit Margin

%



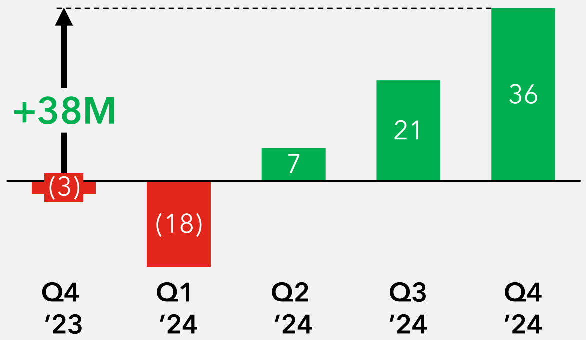
Adj. EBITDA[2] and Net Income

$ in millions



Cash from Operations

$ in millions



Note: Figures may not total due to rounding

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Summary

Earning our right to win by leveraging the Aerogel Technology Platform®, executing, and investing to grow profitably

DEMONSTRATED OPERATING EXECUTION

FY 2024 gross margins grew to 40% driving $13.4 million of net income and $89.9 million of Adj EBITDA

INCREASING FLEXIBILITY TO MEET DEMAND

External manufacturing facility capable of increasing aerogel supply in a capital-light and modular manner

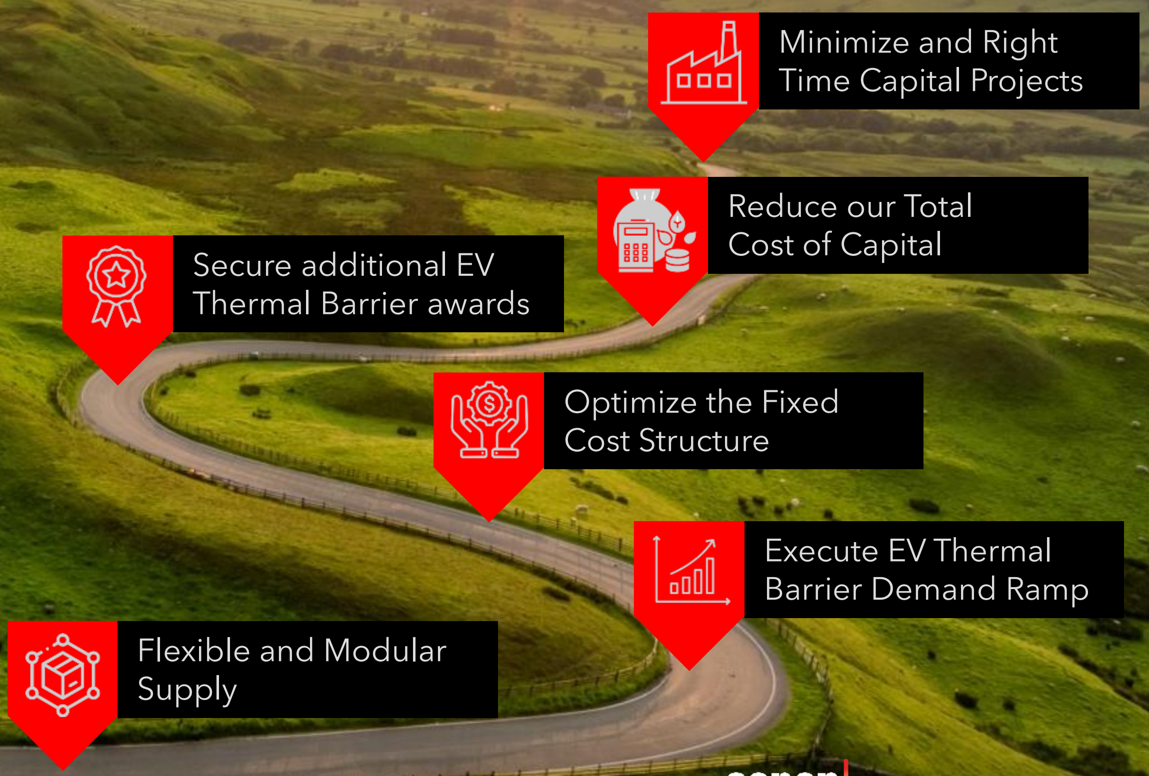
SECURING MORE PYROTHIN® CONTRACTS

PyroThin® award from Volvo Truck, our second commercial vehicle contract and eighth overall OEM award

IMPROVING CAPITAL EFFICIENCY OF OUR MODEL

Significantly reduced capital expenditure plans by demobilizing Statesboro Plant; implemented actions to reduce fixed cost base

The Road to Creating Near Term Value



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Appendix

GAAP to Non-GAAP Reconciliation

($ in millions)	2024 Q1	2024 Q2	2024 Q3	2024 Q4	2023 Q4	FY 2024	FY 2023	Q1-25 Outlook Low	Q1-25 Outlook High
Net Loss	(1.8)	16.8	(13.0)	11.4	(0.5)	13.4	(45.8)	(15.0)	-
Depreciation and Amortization	5.8	6.0	5.3	5.4	4.6	22.5	15.3	5.8	5.8
Stock-based Compensation	4.7	3.0	2.6	2.5	3.2	12.9	11.0	2.7	2.7
Loss on Extinguishment of Debt	-	-	27.5	-	-	27.5	-	-	-
Other (Income) Expense & Income Tax Expense	4.3	3.2	2.9	3.4	1.9	13.7	(3.4)	3.5	3.5
Restructuring Costs	-	-	-	-	-	-	-	3.0	3.0
Adjusted EBITDA	**12.9**	**28.9**	**25.4**	**22.7**	**9.1**	**89.9**	**(22.9)**	**-**	**15.0**

Note: Figures may not total due to rounding

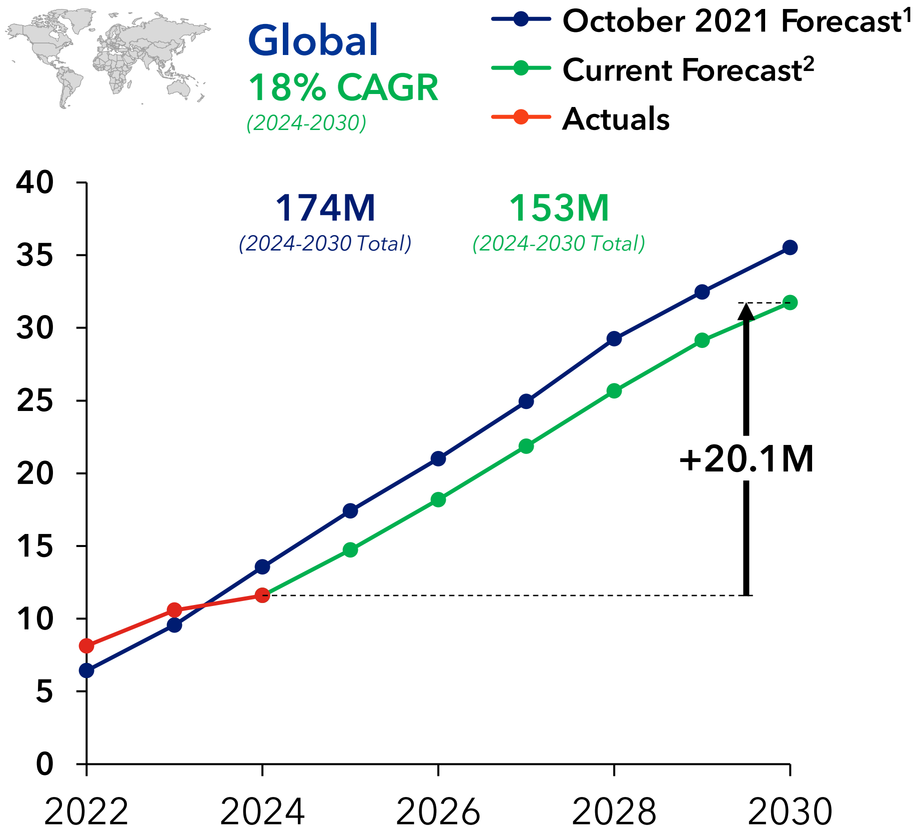
Long-term EV Production Trends

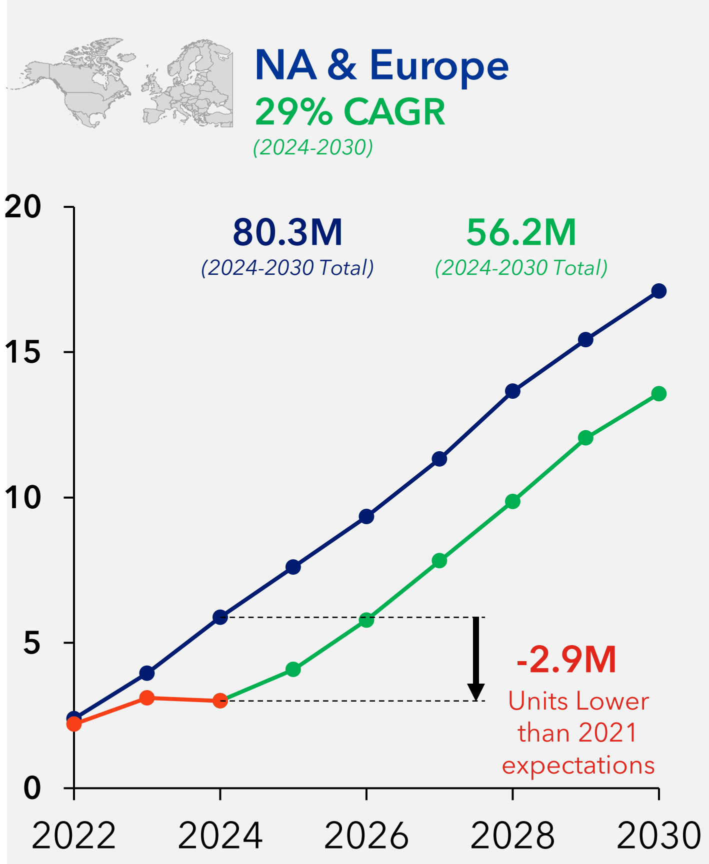
Global EV production is still projected to increase at a 18% CAGR through 2030; 29% CAGR in North America & Europe

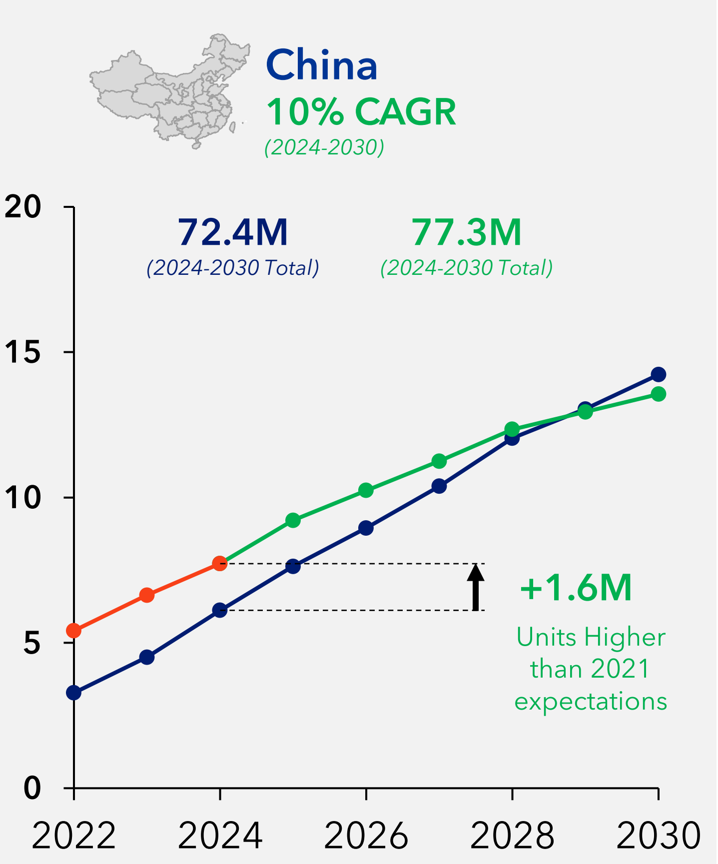
Global EV production outlook expected to increase at 18% CAGR through 2030

…Europe and North America saw BEV production decrease in 2024; China market significantly ahead of 2021 expectations

Electric Vehicle (EV) light vehicle production forecast
Vehicles in Millions







1- IHS October 2021 vehicle production forecast
2 - IHS February 2025 vehicle production forecast

US EV Market Update – New Entrants Gaining Share

GM and Honda were ~19% of all electric vehicles sold in the United States in Q4 2024

GM Ultium and Honda are continuing to capture their fair share in a competitive market with new models

US EV Sales Top Gainers/Losers[1]
(US Q4 2023 vs Q4 2024 change in EV sales in thousands)

	Change (k)	Q4 2024 Sales (k)	Q4 % Change	Q4 '24 EV Mkt Share %	Q4 '23 EV Mkt Share %	Q4 Mkt Share Change	C2C Thermal Barrier?[3]
gm ultium cells	37	43.9	535%	12.6%	2.2%	10.5%	PyroThin
HONDA	23	23.2	N/A	6.7%	0.0%	6.7%	PyroThin
HYUNDAI KIA	8	32.5	34%	9.3%	7.7%	1.7%	None
Ford	4	30.2	16%	8.7%	8.2%	0.5%	None
NISSAN	3	8.5	67%	2.5%	1.6%	0.8%	None
RIVIAN	-3	11.0	-19%	3.2%	4.3%	-1.1%	None
Mercedes-Benz	-4	9.3	-31%	2.7%	4.3%	-1.6%	None
Volkswagen	-9	1.8	-83%	0.5%	3.4%	-2.8%	None
gm Bolt EV / EUV	-13	0.0	-100%	0.0%	4.0%	-4.0%	None
TESLA	-17	149.2	-10%	42.9%	52.6%	-9.7%	None
Others[2]	1	38.0	2%	10.9%	11.8%	-0.9%	
Total	31	347.6	10%	100%	100%	-	

1- Wards Intelligence February 2025 Sales Data
2- Others include Porsche, Stellantis, Toyota, Lucid, Volvo, Subaru, Mazda, JLR, BMW, Audi
3- Munro & Associates Battery Teardowns, A2MAC and Commercial Discussions

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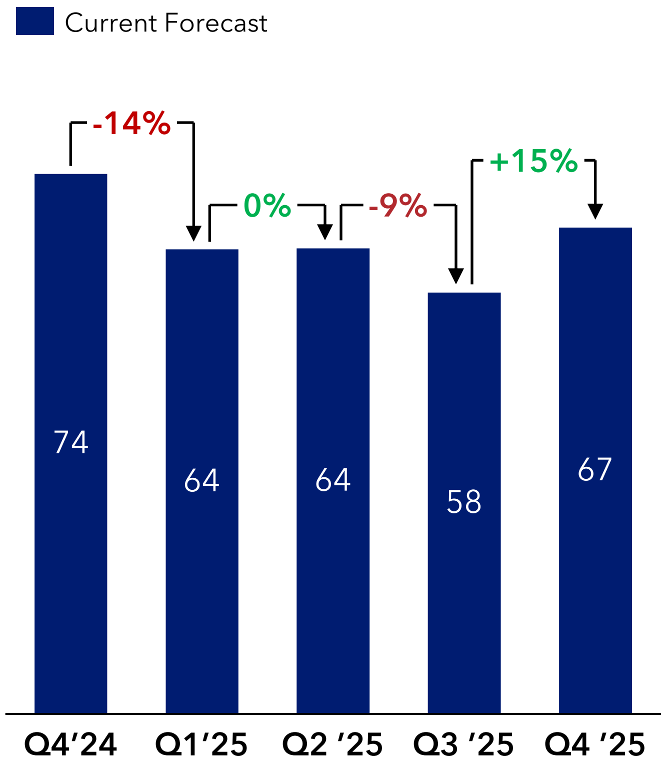
Demand Overview - GM IHS EV Forecast

5% projected volume reduction in 2025 to more closely resemble sales rates; growth expected in 2026/2027

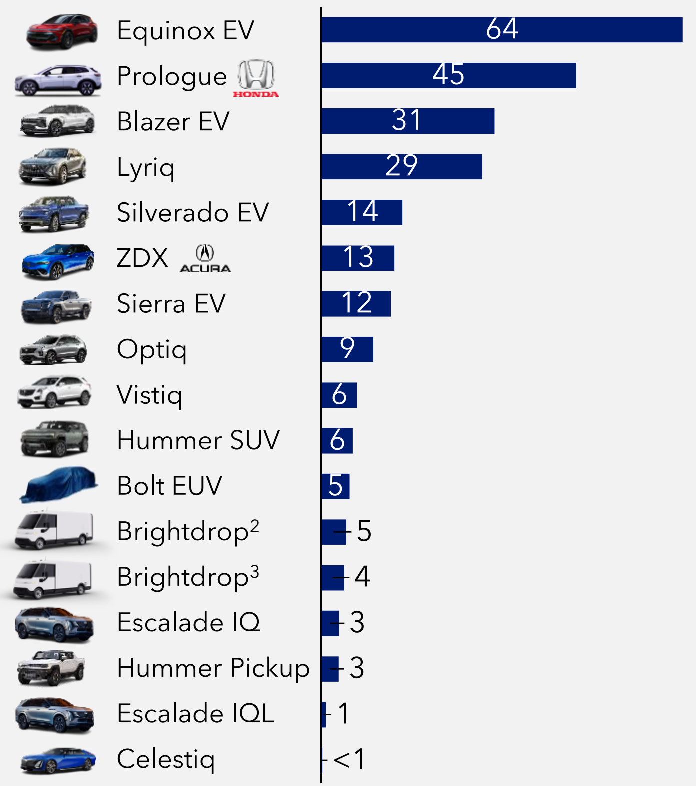
Q1 reduction expected as OEM reaches inventory targets

GM NA EV Production Forecast[1]
(Vehicles in thousands)



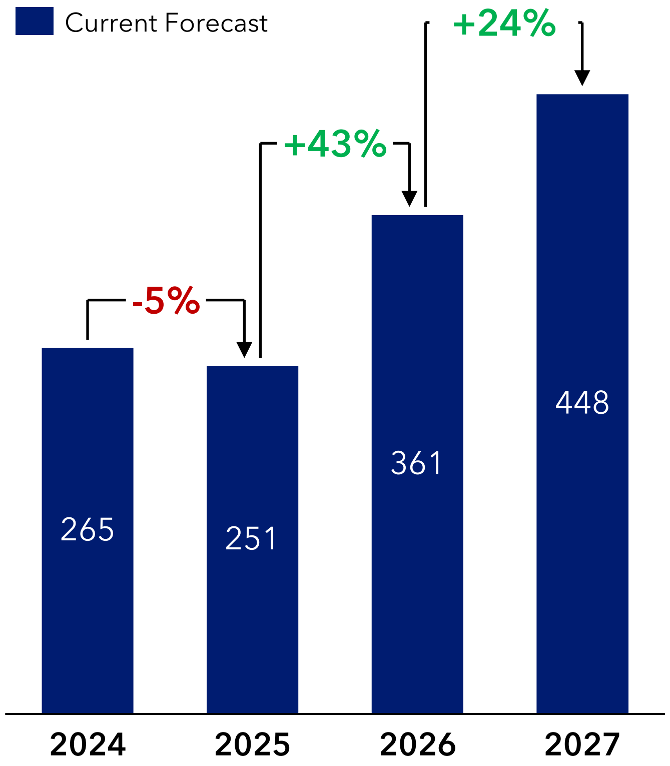
2025 volumes supported by 17 unique nameplates

2025 GM NA EV Production Forecast[1]
(Vehicles in thousands)



2025 with 5% reduction, while long-term outlook remains strong

GM NA EV Production Forecast[1]
(Vehicles in thousands)



1- IHS February 2025 vehicle production forecast; 2025/2026 includes reintroduced Chevy Bolt
2. Brightdrop Zevo 600
3. Brightdrop Zevo 400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